Resultados Validación: Modelo Optimized vs Random

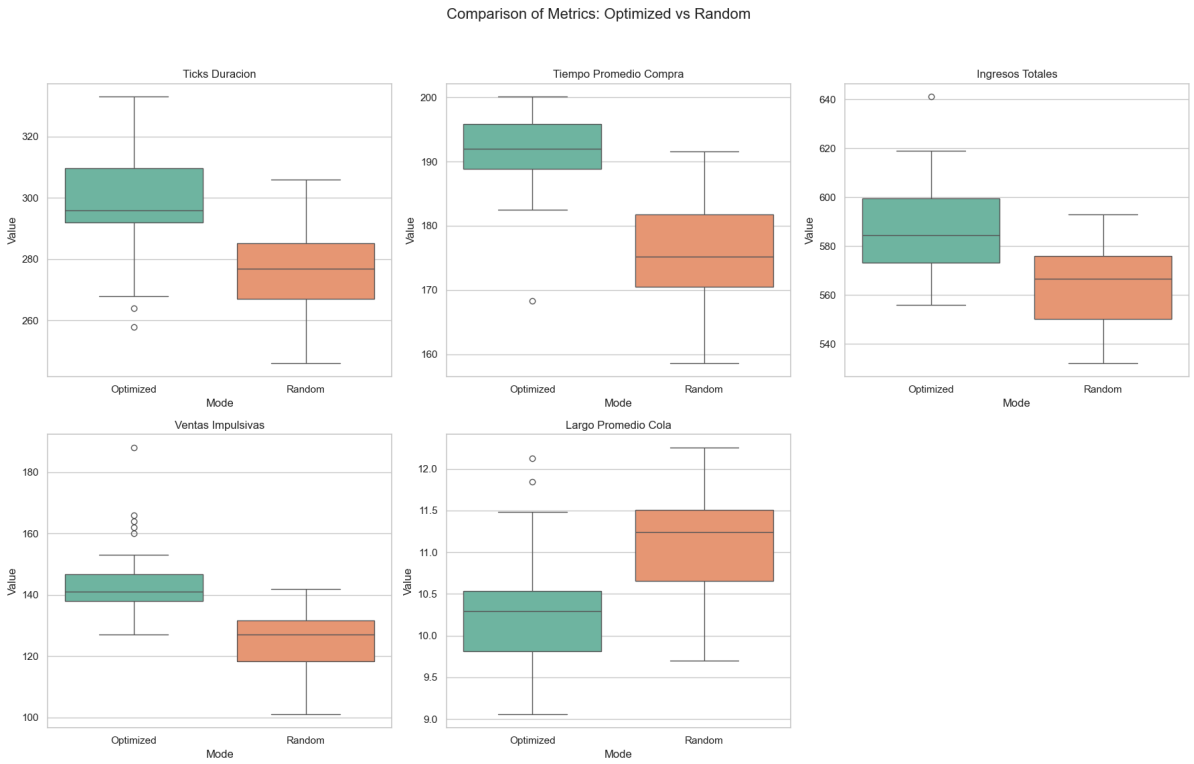
Modo	Episodio	Reorganizaciones	Ticks_Duracion	Tiempo_Promedio_Compra	Ingresos_Totales	Ventas_Impulsivas	Largo_Promedio_Cola
Optimized	1	1	253	167.969697	658	223	12.29249012
Optimized	2	2	291	192.2626263	701	250	10.33676976
Optimized	3	3	346	201.8989899	698	239	9.046242775
Optimized	4	4	280	188.979798	663	229	10.74642857
Optimized	5	5	304	181.4242424	614	195	9.717105263
Optimized	6	6	298	192.6060606	655	223	10.35234899
Optimized	7	7	306	187.0606061	665	229	10.11111111
Optimized	8	8	300	190.5959596	681	233	10.25666667
Optimized	9	9	298	187.0505051	656	221	10.16778523
Optimized	10	10	276	186.1818182	683	248	10.69927536
Optimized	11	11	306	190.0707071	670	231	9.836601307
Optimized	12	12	313	194.5151515	706	251	9.654952077
Optimized	13	13	294	191.2626263	680	228	10.46258503
Optimized	14	14	258	188.979798	692	249	11.65891473
Optimized	15	15	262	189.020202	667	224	11.55725191
Optimized	16	16	258	187.6767677	662	231	12.06589147
Optimized	17	17	309	190.969697	670	234	10.60517799
Optimized	18	18	287	193.010101	637	212	10.84320557
Optimized	19	19	303	185.8585859	655	227	10.08580858
Optimized	20	20	266	190.9292929	682	237	11.38345865
Optimized	21	20	311	194.2020202	703	251	9.823151125
Optimized	22	20	291	188.8181818	673	242	10.9347079
Optimized	23	20	252	188.7575758	685	248	11.78174603
Optimized	24	20	287	195.1919192	705	251	10.51567944
Optimized	25	20	289	189.1313131	672	238	10.60553633
Optimized	26	20	283	185.6767677	682	239	11.28975265
Optimized	27	20	304	192.5151515	692	247	10.19078947
Optimized	28	20	285	195.4545455	715	266	10.63157895
Optimized	29	20	271	193.2222222	698	246	11.65313653
Optimized	30	20	299	187.9292929	653	225	10.30769231
Mean Score		13.66	289.33	189.64	675.76	235.56	10.65

Modo	Episodio	Reorganizaciones	Ticks_Duracion	Tiempo_Promedio_Compra	Ingresos_Totales	Ventas_Impulsiva	Largo_Promedio_Cola
Random	1	0	286	182.3030303	572	132	10.58041958
Random	2	0	278	173.4343434	540	106	11.06834532
Random	3	0	296	176.6161616	570	135	10.43918919
Random	4	0	276	182.4444444	558	116	11.35507246
Random	5	0	293	191.1818182	585	135	10.61433447
Random	6	0	267	170.2121212	560	122	11.41947566
Random	7	0	276	180.3232323	590	142	11.27173913
Random	8	0	278	183.5858586	544	117	11.20863309
Random	9	0	278	191.5454545	540	111	11.51438849
Random	10	0	265	168.4646465	532	101	11.48301887
Random	11	0	261	169.8383838	577	142	11.94252874
Random	12	0	266	164.9494949	548	127	11.10526316
Random	13	0	246	164.1313131	561	130	12.00813008
Random	14	0	269	178.3131313	590	138	11.51672862
Random	15	0	303	184.2020202	567	122	10.55775578
Random	16	0	279	178.6464646	546	115	10.93189964
Random	17	0	305	179.5353535	585	127	9.806557377
Random	18	0	261	187.4444444	560	128	11.86590038
Random	19	0	271	171.2828283	574	128	11.4797048
Random	20	0	265	172.1111111	583	132	11.77358491
Random	21	0	283	173.1515152	566	123	10.77385159
Random	22	0	258	171.5858586	593	139	12.25193798
Random	23	0	297	173.4141414	546	110	10.12794613
Random	24	0	271	162.8484848	576	129	11.29889299
Random	25	0	283	174.9292929	568	124	11.15194346
Random	26	0	271	175.5151515	568	122	11.70848708
Random	27	0	267	158.5555556	534	115	11.4494382
Random	28	0	280	179.8282828	564	131	10.95
Random	29	0	300	186.9292929	557	126	10.23666667
Random	30	0	306	166.2828283	576	131	9.702614379
Mean Score			277.83	175.78	564.33	125.2	11.11

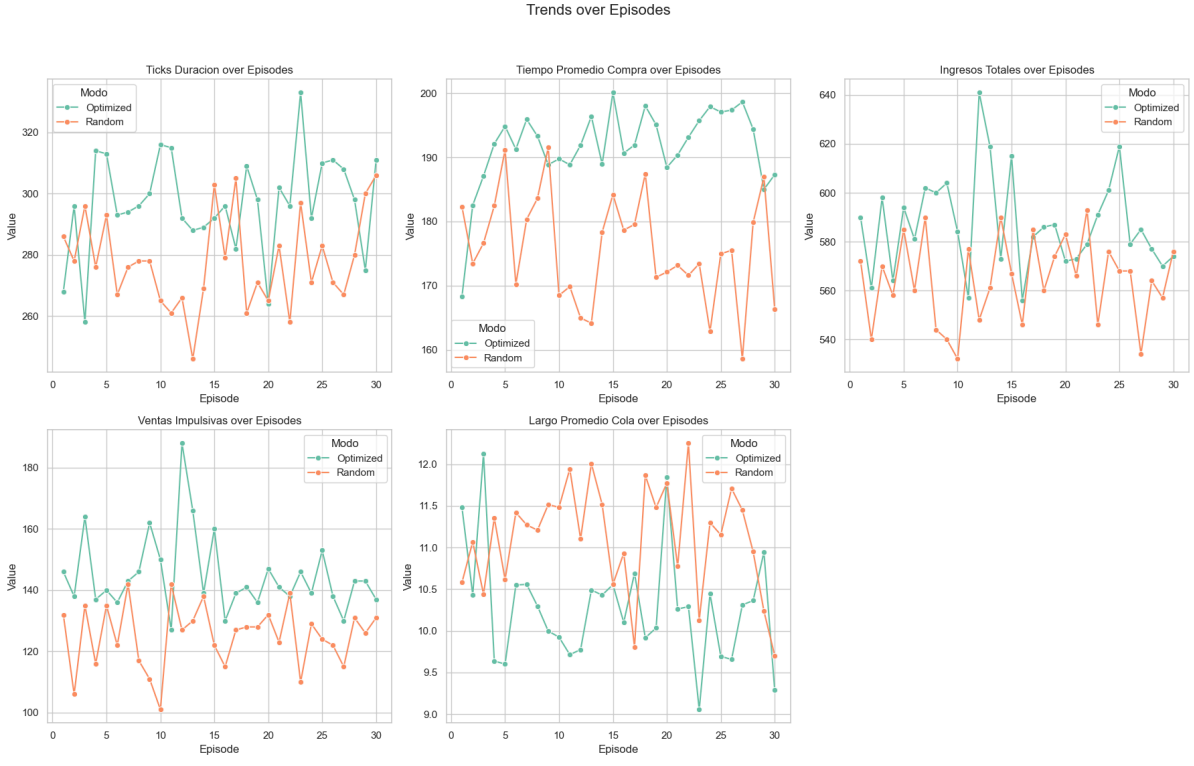
Experimentos realizados con num-customers = 100; rearrangement-intervals=150; impulse-buy-prob= 0.10;
Rearrangement Policy=0 (para Random) o 2 (para Optimized); popular-item-prob= 0.90.
20 iteraciones para el reordenamiento y 10 iteraciones de testing. 30 iteraciones en total.

Paper usado para la validación: ***Evaluating customer satisfaction and store revenue using agent-based modeling of grocery store layouts***

- by Livia Tengelin & Sebastian Zulj



Boxplots Optimized vs Random (General)



Line Graph para cada iteración individual.